

Voice-guided ATMs for the visually-challenged

"India has one of the largest visually-impaired populations in the world. The Reserve Bank of India, in its circulars in 2008 and 2009, stated that all banking services including ATM cards should be offered to customers with disabilities, and without any discrimination. NCR Corporation—India's largest ATM manufacturer—wanted to reach out to this population and give them a chance to transact independently. Through research, NCR understood the banking challenges that a visually-impaired person usually faces and thereafter developed the Talking ATM—a customised ATM that will benefit such customers while also maintaining the safety of the transaction. We've been in constant discussions with the Xavier's Resource Centre for the Visually Challenged (XRCVC) to identify the needs of visually-impaired people before launching the Talking ATMs," says Navroze Dastur, managing director, NCR India.

The Talking ATM is also attractive to banks because they don't have to issue any special cards—a regular ATM card will work on the Talking ATM too!

NCR's Talking ATM uses a text-to-speech engine, which allows voicing-out the text on the screen in multiple languages for consumer convenience. This machine is incorporated with unique software and hardware features that ensure that a person with a disability can operate the machine on his own while maintaining the safety of the transaction. It comprises accessible keypads and Braille stickers too.

When a visually-challenged person attaches his headphone set to this ATM, he can hear the instructions, which enables him to fill-in the required data using the numeric keypad. Apart from reading aloud screen messages, the machine provides complete orientation, making it easy for the customer to use the machine.

An important security feature of this ATM is that it provides the person an option to blank out the screen as a safety mechanism to avoid shoulder-surfing by bystanders.

The system is completely Access for



Union Bank Talking ATM for visually impaired
(Courtesy: www.addressofwealth.com)

All (AFA) compliant, and also includes ramps for those with other physical disabilities. Banks can also make their existing ATM network AFA-compliant by customising the ATM software stack and upgrading the hardware configuration of the ATM fleet.

NCR has showcased its 'Talking ATM' at different workshops conducted by XRCVC over the last couple of years. "In June 2012, NCR got the order of a hundred Talking ATMs from the Union Bank of India for deployment in passport offices and other UBI branches across India. This shows that Indian banks and financial institutions are quickly realising the need to adapt self-service technologies to include millions of differently-abled people into the financial stream. NCR is committed to helping these institutions by providing innovative technologies, most of which are conceptualised, created and manufactured in India," says Dastur.

These ATMs can be procured from, deployed and serviced by NCR. Any financial institution can upgrade its existing ATM to a Talking ATM by making minor changes in the hardware and software configuration, which is not very expensive. Thus the price of the ATM depends on the features added and the scale of the ATM roll-out.

Touchscreens and virtual reality open new windows for autistic children

Autism experts feel that touchscreen technology is a boon to children with Autism Spectrum Disorder (ASD), as it helps expand their skills and explore new worlds all at the touch or even

bang of a button! Of late, we saw a lot of groups developing smartphone-based apps and devices for young children with autism.

Access4Kids, for example, is a wireless input device for children with fine motor impairments. Developed by Ayanna Howard, Giancarlo Valentin and Hae Won Park from Georgia Tech, the device may be worn on the forearm or placed on a wheelchair arm. It includes three force-sensitive resistors that measure pressure from physical movements such as rough hits or swipes, and converts them into fine-motor gestures that translate to touchscreen commands.

Coupled with an Android-powered tablet computer and a special software stack, it becomes a holistic therapeutic system for children lacking upper-body control. The open source software library developed by the team is a critical component of the system, as it enables the kids to access popular applications like Facebook and YouTube as well as custom-made apps for therapy and education. The first prototype has been successful, and clinical trials are likely to begin this year.

This year's *Consumer Electronics Show* also showcased several products with features that could be used by autistic children too.

One interesting example is Big Bird's Words, which could help parents of special children to convert nightmarish shopping trips into fun-filled learning experiences. The outcome of collaboration between Sesame Street and Qualcomm, Big Bird's Words lets kids use their parents' phone to scan the world around them for printed words. Big Bird then helps them learn to read by sounding out the first letter. For example, when you go shopping, you could get the kids to match the items on your list (preloaded into your mobile as images) with the boards on the shelves. The game starts by asking the users to pick a word with an associated image. The users then have to find that word in the real world by pointing the camera of the device at the text.